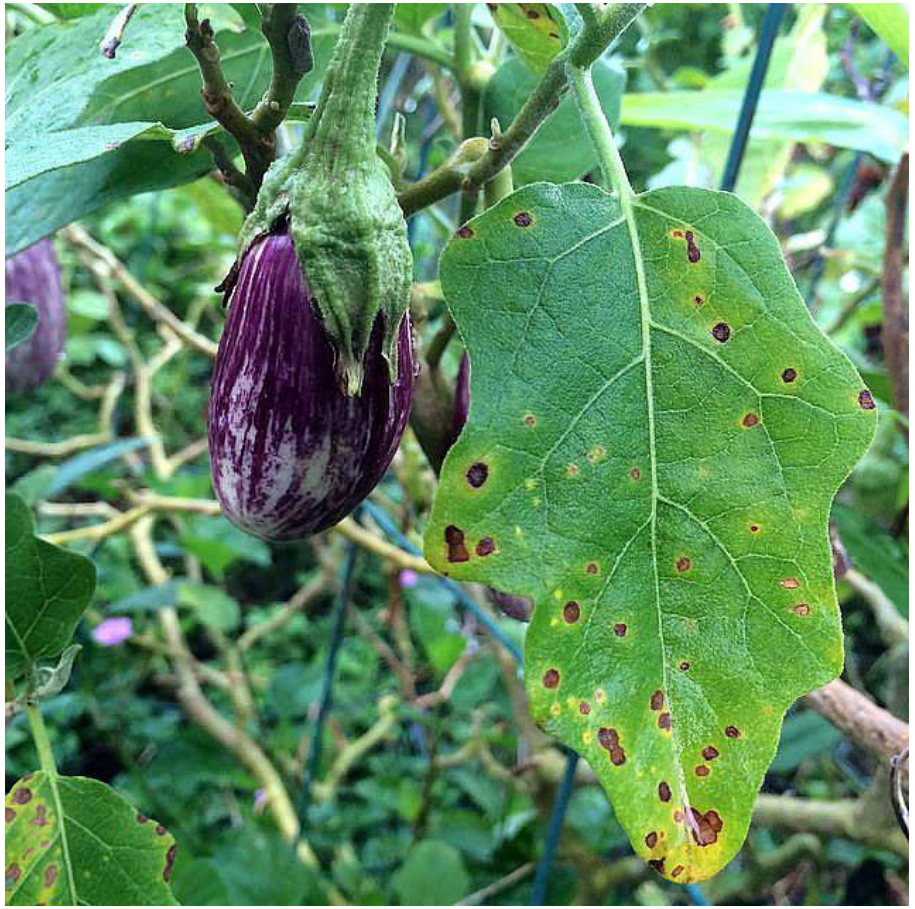
# Brinjal diseases

1. Damping off - *Pythium aphanidermatum*, *P. indicum*, , *Phytophthora parasitica*, *Rhizoctonia solani* and *Sclerotium rolfsii*.
2. Cercospora leaf spot - *Cerospora solanigena*, *C.melongenae*, *C. egenulae*
3. Alternaria leaf spot - *Alternaria melongenae*, *A. solani*
4. Phomopsis blight and fruit rot - *Phomopsis vexans*
5. Collar Rot - *Sclerotium rolfsii*
6. Verticillium wilt - *Verticillium dahliae*
7. Bacterial wilt - *Ralstonia solanacearum*
8. Little leaf - Phytoplasma

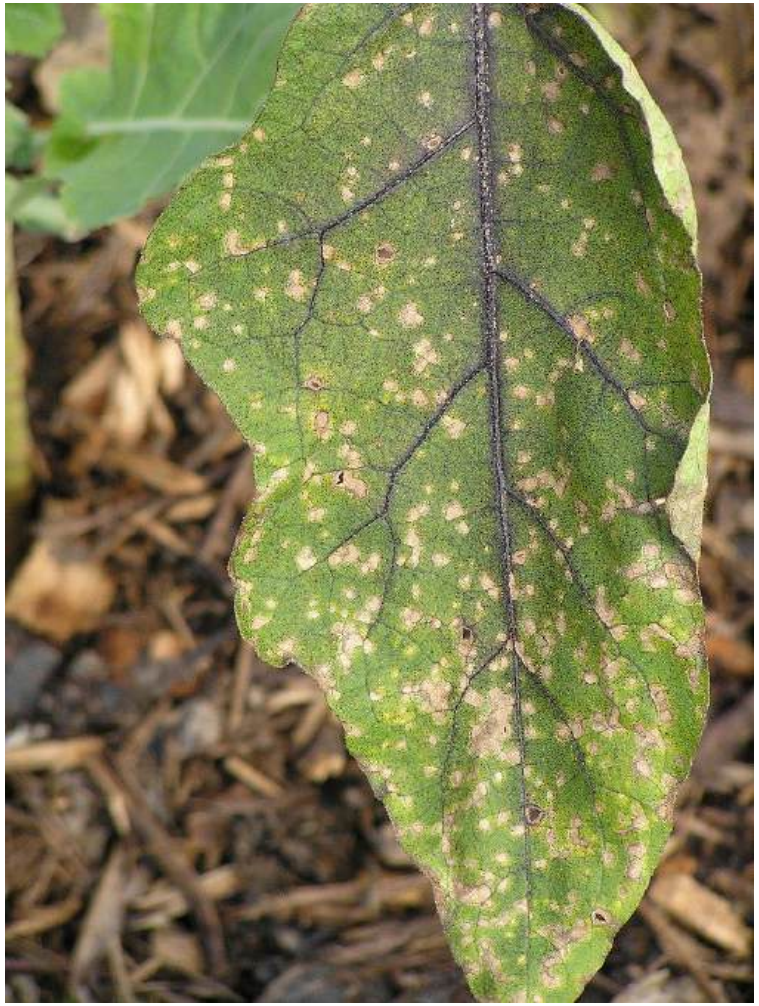
**Damping off - *Pythium aphanidermatum, P. indicum, , Phytophthora parasitica, Rhizoctonia solani* and *Sclerotium rolfsii.***

Symptoms, pathogen, disease cycle, epidemiology and management **similar to tomato.**

# *Cercospora leaf spot –Cerospora solanigena, C.melongenae, C. egenulae*



# *Cercospora leaf spot –Cerospora solanigena, C.melongenae, C. egenulae*



# *Cercospora leaf spot –Cerospora solanigena, C.melongenae, C. egenulae*

Several leaf spot disease affects this crop but the leaf spots caused by *Cercospora* species are important which under high humidity conditions cause considerable yield losses.

## Symptoms

### *Cercospora melongenae*

- The symptoms of the disease appear as circular to irregular leaf spots, which are **usually large and brown to grayish brown in colour.**
- Later spots coalesce together and affected leaves fall down prematurely and **sometimes fruits may rot.**

### *Cercospora solanigena*

- Leaf spots **amphigenous circular** to irregular later coalescing, necrotic, **dark brown in center and light brown in margin, 0.5-2 mm wide.**

# *Cercospora leaf spot –Cerospora solanigena, C.melongenae, C. egenulae*

**Symptoms**

***Cercospora egenulae*:**

- Early infection noticed on the lower leaves which gradually spreads upwards with high humidity and rainfall.
- Early symptoms start as **minute water-soaked specks**, which gradually enlarge to become **circular to angular leaf spots of 2-6 mm diameter.**
- These spots often **coalesce to form bigger spots having a papery pale yellow** necrotic region in the center.
- The necrotic region in the center disintegrates bearing **a shot hole**
- Infected leaves turn yellow, droop and fall prematurely resulting in quick defoliation of the entire plant.

# *Cercospora leaf spot –Cerospora solanigena, C.melongenae, C. egenulae*

**Disease cycle and epidemiology:**

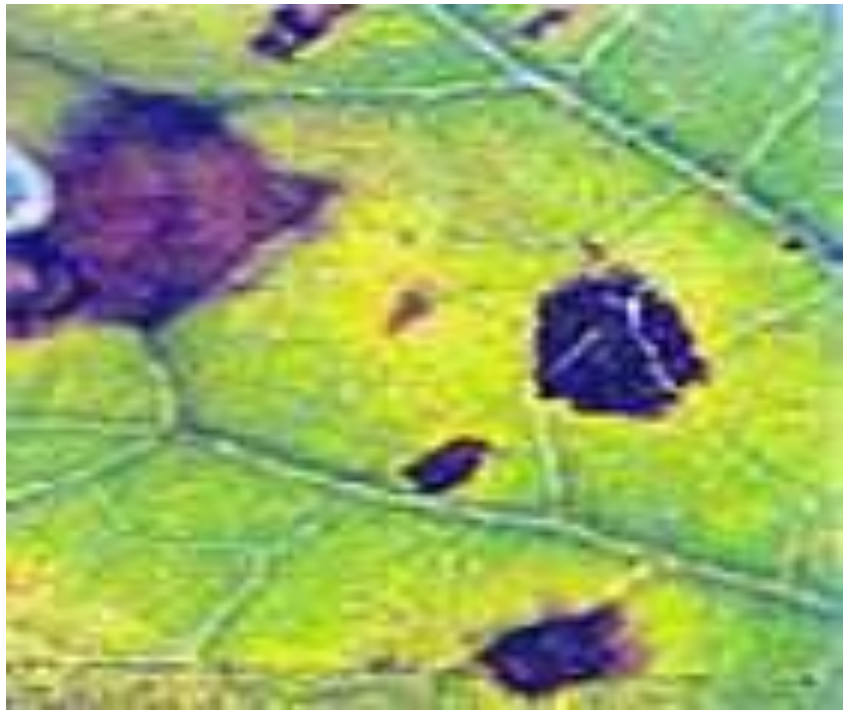
- The fungus survives from one season to another in infected plant debris in soil and also in infected seeds wherever fruit rot occurs.
- **Warm days and cool nights are ideal for the infection.**
- The disease is favoured by high humidity and heavy persistent dews.
- **Water droplets must be present for spore germination and germ-tube penetration.**
- Moist wind, irrigation water and insects help in local transmission of the pathogen.

## *Cercospora leaf spot –Cerospora solanigena, C.melongenae, C. egenulae*

**Management:**

- Cultural practices like destruction of crop debris, crop rotation, use of disease free seeds and wider plant spacing should be followed to reduce the primary inoculum of the pathogen in field.
- With the initiation of the disease spraying with zineb (0.25%), carbendazim (0.1%) or thiophanate methyl (0.1%) and repeat at 10 to 14 days interval.

# Alternaria leaf spot-*Alternaria melongenae, A. solani*



# Alternaria leaf spot-*Alternaria melongenae, A. solani*

Disease initiates late in **October to early November**

Severe during **November and December**

## Symptoms

**On leaves**

**Brown & irregular spots with concentric rings**.

Several spots coalesce to form bigger necrotic patches.

**On fruits**

***A. melongenae* infects the fruits with large deep sunken spots**.

The infection on the fruit causes large deep sunken spots

The infected fruits turn yellow & drop pre-maturely.

# Alternaria leaf spot-*Alternaria melongenae, A. solani*

## Management

- Seed treatment with thiram or captan
- **Pant samrat** variety is resistant.
- **Hot water treatment @50$^{O}$C/3-4mins.**

# Phomopsis blight and fruit rot of brinjal -*Phomopsis vexans* (PS:*Diaporthe vexans*)

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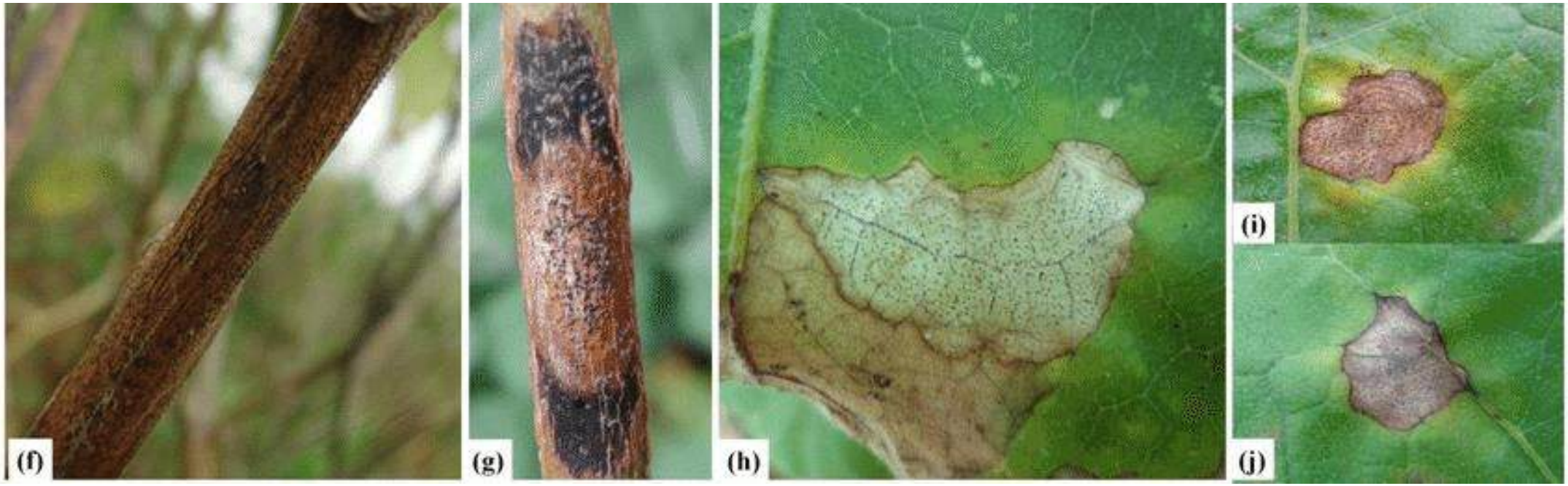
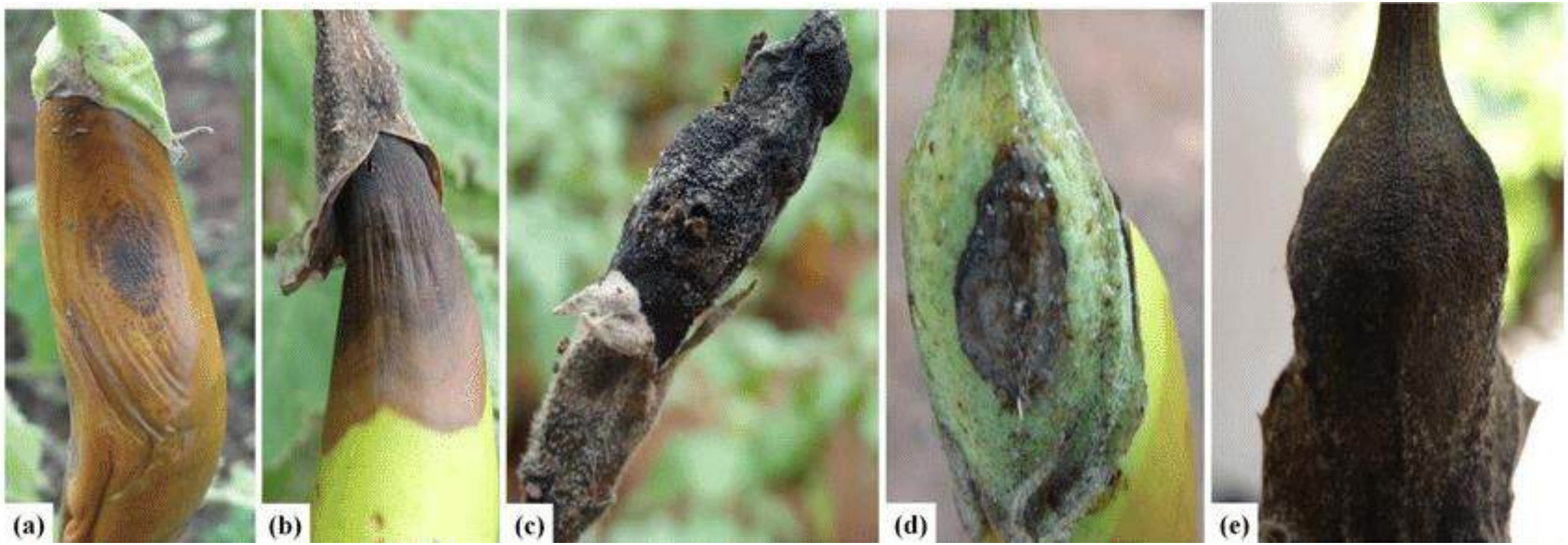


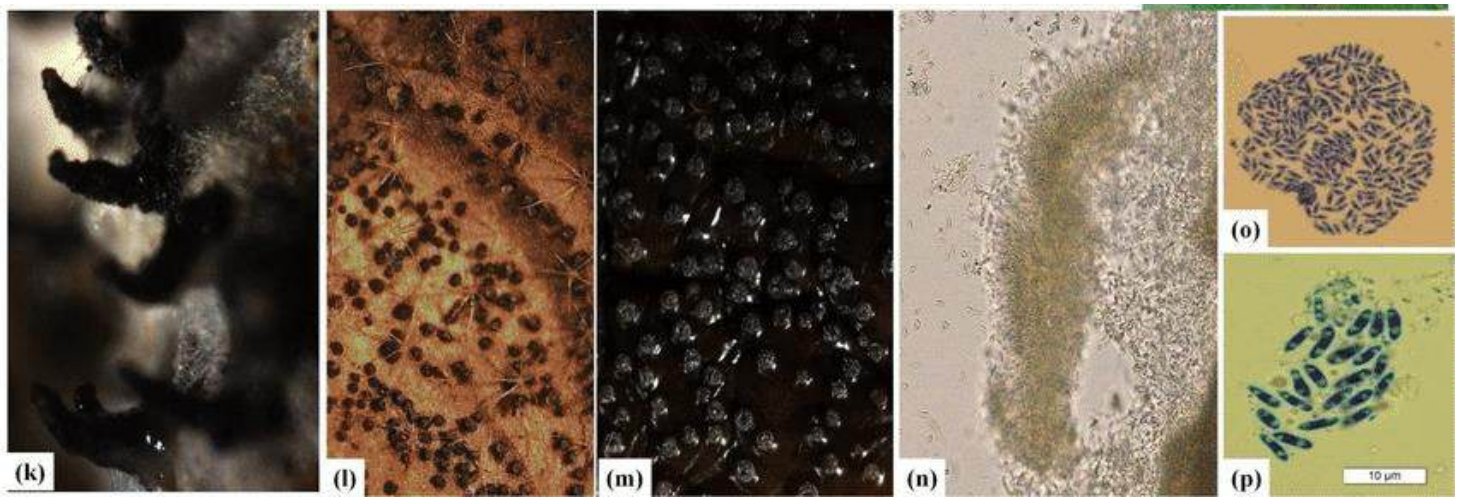

Disease symptoms caused by *Phomopsis vexans* stem blight (f, g) and leaf blight (h, I, j) on severely infected plants

# Phomopsis blight and fruit rot of brinjal -*Phomopsis vexans* (PS:*Diaporthe vexans*)




Disease symptoms caused by *Phomopsis vexans*. (a,b) (fruit rot of brinjal, (c) mummified fruit, (d, e) blight on calyx

# Phomopsis blight and fruit rot of brinjal -*Phomopsis vexans* (PS:*Diaporthe vexans*)




Pycnidia on affected leaf and fruit, (n) pycnidia (100 µm), (o, p) alpha conidia.

# Phomopsis blight and fruit rot of brinjal -*Phomopsis vexans* (PS:*Diaporthe vexans*)

# Phomopsis blight and fruit rot of brinjal -*Phomopsis vexans* (PS:*Diaporthe vexans*)

- Serious disease of brinjal- **Phomopsis leaf blight, stem blight, Phomopsis rot, fruit rot, brown spot and tip over of brinjal**
- In India, it was reported in 1935 in Gujarat
- Attacks foliage and fruits, fruit rot phase is more destructive with substantial loss in field and transit
- Reduces the yield by about 40 – 70%

# Phomopsis blight and fruit rot of brinjal -*Phomopsis vexans* (PS:*Diaporthe vexans*)

## Symptoms:

The plants are attacked at all stages of growth with the following symptoms

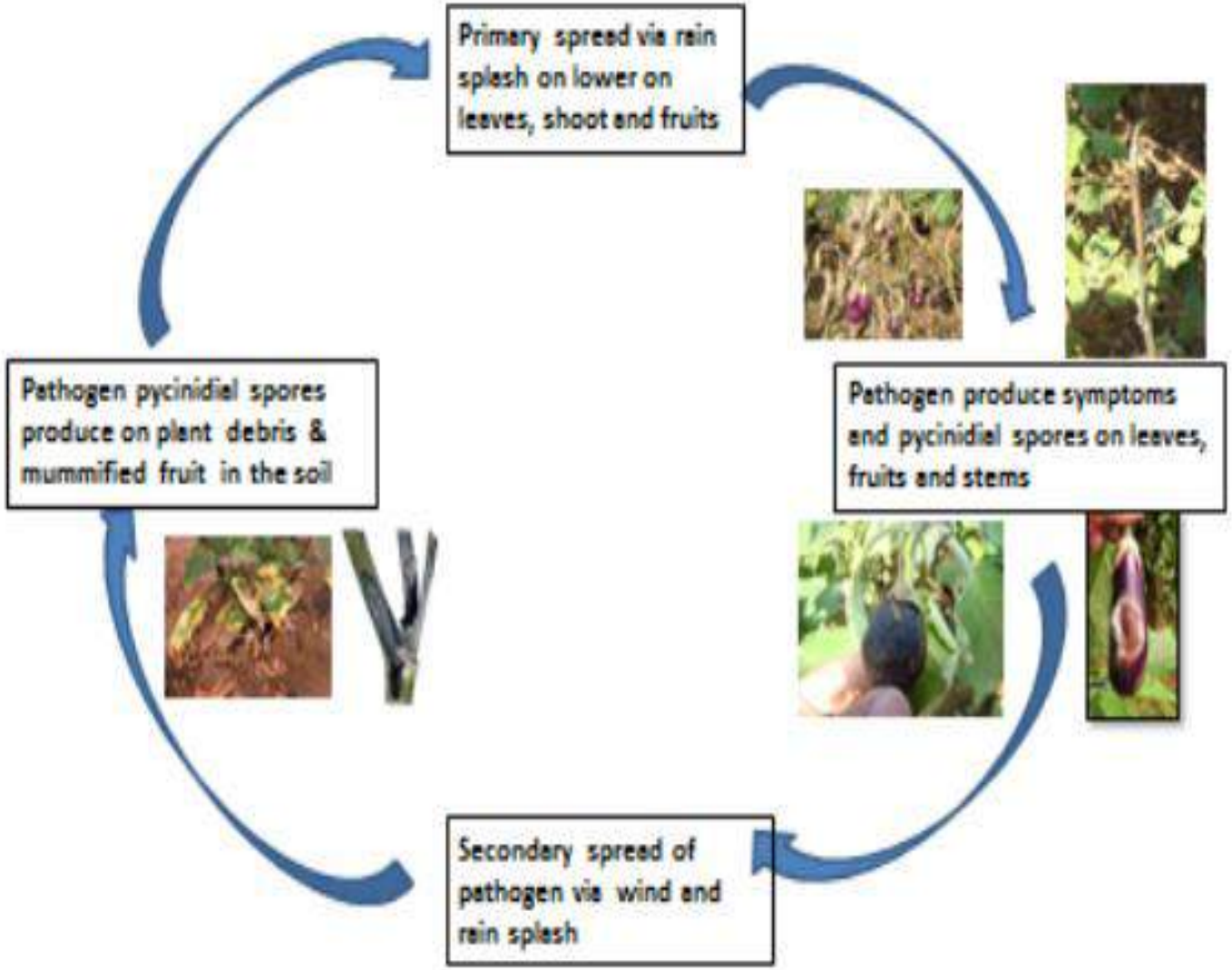
- Damping off
- Stem canker
- Leaf blight
- Fruit rot
- In nursery bed, it causes damping off which results from the infection of the seedlings at the collar portion just above the soil line.
- After transplanting, **circular grey spots with light coloured centers appear on the leaves**
- **Lesions on petioles and lower part of the mid rib results in death of the leaf**
- In later stages, the lighter portion is studded with numerous black pycnidia.
- Affected leaves turn yellow and fall down pre-maturely.

# Phomopsis blight and fruit rot of brinjal -*Phomopsis vexans* (PS:*Diaporthe vexans*)

**Symptoms:**

- Auxillary buds often killed
- **On stem, at the base** characteristic constrictions leads to **canker development and toppling of the plants**.
- **Breaking of the stem** at the point of infection due to strong winds
- **On fruits, the disease manifests as pale sunken spots,** which later enlarge to form rotten areas with **concentric yellow and brown rings**
- A large number of dot like pycnidia also develop on such spots.
- **If infection of fruits takes place through calyx, which is the most infective part,** the whole fruit becomes mummified due to dry rot.

# Phomopsis blight and fruit rot of brinjal -*Phomopsis vexans* (PS:*Diaporthe vexans*)



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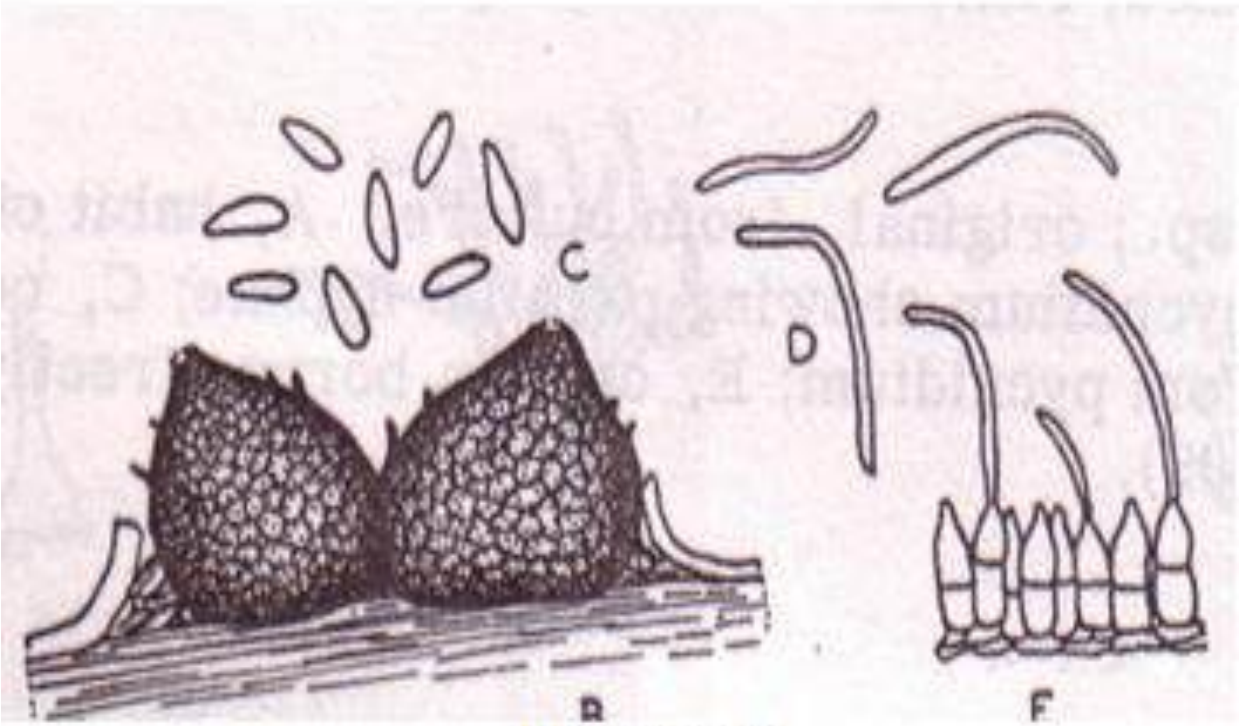



*Phomopsis*

# Phomopsis blight and fruit rot of brinjal -*Phomopsis vexans* (PS:*Diaporthe vexans*)

## Disease cycle and epidemiology:

- The pathogen is seed borne and also survives in plant debris both as mycelium and pycnidia
- Seed infections directly lead to diseased seedlings.
- The pycnidiospores are disseminated through rain splashes, irrigation water, agricultural implements and insects.
- **High relative humidity coupled with high temperatures** is favourable for disease development.
- Pathogen grows best at a temperature of **21-32.5°C** with a optimum temperature of **28 -29°C**. Storage rot is maximum at a temperature of 25°C.
- Maximum disease development takes place at about 26°C under wet weather conditions.

# *Phomopsis blight and fruit rot of brinjal -Phomopsis vexans (PS:Diaporthe vexans)*

## Favourable conditions

- *Phomopsis vexans* requires **hot and humid climate** for causing infection and disease development.
- Maximum disease development takes place **at about $26^0$C under wet weather conditions with 55% R.H.**
- **The temperature greater than 40°C** was not favorable for the development of the disease.

# Phomopsis blight and fruit rot of brinjal -*Phomopsis vexans* (PS:*Diaporthe vexans*)

## Management:

- **Hot water treatment of seed at 55°C /15 min or 50°C / 30min**
- Application of ammonical form of nitrogenous fertilizers and increased rate of 30-60 kg/ha
- Rest. Var. Pant Samrat
- Cultural practices like collection and destruction of the diseased plant debris, **crop rotation - 3 years** and use of disease free seeds are recommended to reduce the initial inoculum.

# Phomopsis blight and fruit rot of brinjal -*Phomopsis vexans* (PS:*Diaporthe vexans*)

## Management contd…

- Treat the seed with carbendazim (0.2%) or thiophanate methyl (0.2%). With the initiation of the disease.

- Spraying with carbendazim (0.1%) or combination of mancozeb (0.25%) and carbendazim (0.05%) or copper oxychloride (0.3%) and repeat at 10 to 14 days interval.

- **Aqueous extracts of asoka, neem, maredu, tulasi, *Vinca rosea***

# Collar Rot - *Sclerotium rolfsii*

**Sclerotial growth on collar region**

# Collar Rot - *Sclerotium rolfsii*

# Collar Rot- *Sclerotium rolfsii*

## Symptoms

- The lower part of the stem is affected first.
- Small water soaked lesions
- Constricted at the base - leading to decortications - chief characteristic symptom
- Necrosis of the tissues leads to collapse of the plant.
- Whitish growth of mycelium & brown mustard like sclerotia on the stem near the soil.

# Collar Rot- *Sclerotium rolfsii*

## Disease cycle:

Sclerotial bodies in soil

As saprophyte in soil

## Management

1. Crop rotation with paddy & banana will reduce the intensity.
2. Drenching the plant base with chestnut compound at 3 gm/lt.

# Verticillium wilt- *Verticillium dahliae*

# Verticillium wilt- *Verticillium dahliae*

Vascular streaking within the stem tissue of an eggplant.

# Verticillium wilt- *Verticillium dahliae*

# Verticillium wilt- *Verticillium dahliae*

- Symptoms become more evident with increasing water demand when the **plants aged in fruiting stage under are hot and dry conditions**
- Symptoms are often expressed **on one side of a plant, branch, or leaflets**.
- Symptoms **may not appear until initiation of flowering**.
- Plants may wilt in the afternoon when they are actively transpiring and appear to "recover" in the morning, only to wilt again.
- **Wilting side of a leaf may be smaller** than the unaffected side when pathogen was starting to clog vascular tissue before the leaf was fully expanded, giving the leaf a lopsided appearance.
- Vascular tissue inside the main stem becomes brown which is visible when the stem is cut in half longitudinally.

# Verticillium wilt- *Verticillium dahliae*

## Disease Cycle and Epidemiology

- Verticillium wilt is soilborne, fungus infects **the roots of host plants**
- Overwinters as a survival structure called a microsclerotium can survive for up to 10 years in the soil without host and withstand drought and cold temperatures.
- Ideal conditions for Verticillium wilt include cooler weather (75 to 85 ºF) and **neutral to alkaline soil (pH of 7 to 9).**
- The host root exudates trigger germination microsclerotia in soil.

# Verticillium wilt- *Verticillium dahliae*

## Management

- Avoid application of **higher doses of nitrogen**
- Rotation with non host crops
- Mulching with polythene sheet
- Soil fumigation
- Drenching of the soil around the plants with carbendazim 0.1% is recommended

# Bacterial wilt - *Ralstonia solanacearum*

Symptoms, pathogen, disease cycle, epidemiology and management **similar to tomato.**

# **Bacterial wilt-** *Ralstonia solanacearum*

# **Bacterial wilt-** *Ralstonia solanacearum*

- Bacterial wilt is one of the most devastating disease in solanaceous plants.
- It is more serious in warm temperate, subtropical and tropical regions of the world.
- The disease is also called southern bacterial wilt, solanaceous wilt, southern bacterial blight and by many other common names in countries wherever it occurs.

## Symptoms

- Characteristic symptoms of bacterial wilt on most of the hosts are sudden wilting and death of infected plants.
- Petioles of the older leaves of the infected plants droop down without yellowing.

# **Bacterial wilt-** *Ralstonia solanacearum*

## Symptoms

- Leaves show epinasty symptoms accompanied by yellowing and stunting of whole plant.
- There is a typical browning of the vascular tissues of roots and stems
- Bacterial polysaccharides mechanically block the vascular system, which checks the translocation of water and minerals resulting in wilting of plants.
- Excessive development of adventitious roots may also take place in tomato.
- Cross sectioned part of the affected root or stem yield whitish milky bacterial ooze.

# **Bacterial wilt-** *Ralstonia solanacearum*

**Pathogen:**

- The disease is caused by *Ralstonia solanacearum*
- The bacterium is gram negative, rod shaped, frequently occurs in pairs, motile with 1 to 4 polar flagella and measures 0.5-0.7 x 1.5 – 2.5 µm.
- **Five races (race1, 2, 3, 4, 5) and four biovars (biovars I, II, III and IV) are known to occur in different parts of the world.**
- In India **race 1 and 3 and biovar II, III and IV are prevalent**.

# Bacterial wilt - *Ralstonia solanacearum*

- Pathogen infects more than 200 plant species in 40 families
- Bacterium survives in plant debris and in moist soil (longer periods)
- It colonizes the xylem vessels
- Also invades intercellular spaces of parenchyma cells in the pith and cortex region
- Disintegrates the cell walls results in pockets filled with bacterial cells and polysaccharides and cellular debris
- Bactria released into the soil from the infected roots and decaying plant material

# **Bacterial wilt-** *Ralstonia solanacearum*

## Disease cycle and epidemiology:

- The pathogen is **both soil and seed borne in nature** and **overwinters in infected plant parts, in potato tubers, in wild host plants and solanaceous weeds** and can survive at least for **2 years in the absence of any host**.
- Injured and decaying infected tissues release bacteria in the soil which enter plants through wounds occurring in roots due to cultural practices, nematodes and insects.
- **Relatively high soil moisture and soil temperature favour the disease.**

# **Bacterial wilt-** *Ralstonia solanacearum*

## Management

- Control of bacterial wilt in infested soil is very difficult as all the commercial cultivars are susceptible and no chemical control is available.
- Follow long crop rotation with non-solanaceous crops.
- Avoid the movement of water from infected plant to healthy plants.
- Shifting of date of transplanting to avoid period of high temperature, heavy rainfall or both.
- Destruction of weeds and volunteer plants
- Disinfection of tools and implements
- Avoid movement of workers in infected areas
- Soil sterilization / steam sterilization
- **Res. Var. Arka Needhi, Arka Neelkanth and Arka Keshav**

# **Bacterial wilt-** *Ralstonia solanacearum*

**Management contd…**

- **Green manuring or biofumigation** with *Brassica* spp. may reduce the bacterial inoculum in soil.
- **Bacterial bioagents** - *Pseudomonas fluorescens*, *P. glumae*, *P.* cepacia and *Bacillus* spp.
- Application of bleaching powder (15 kg/ ha) has also been found effective.
- The disease can also be controlled **effectively if dazomet application is combined with soil solarization.**
- Seedling dip in Streptocycline (100 ppm) for 30 minutes is also effective to some extent.

# Little leaf of brinjal – Phytoplasma

# Little leaf of brinjal – Phytoplasma

- In India it was first reported from Coimbatore
- The pathogen perenneates in weed hosts, *Datura fastuosa*, *D. stramonium*, *Vinca rosea*, *Catharanthus roseus*, *Argemone mexicana*, chilli, tomato and tobacco.
- **Jassid – *Hishimonas (Cestius) phycitis*.**
- The disease is also transmitted by ***Emposca devastans***, however, **it is a less efficient vector.**
- The varieties **pusa purple long and selection T** are highly susceptible.

# Little leaf of brinjal – Phytoplasma

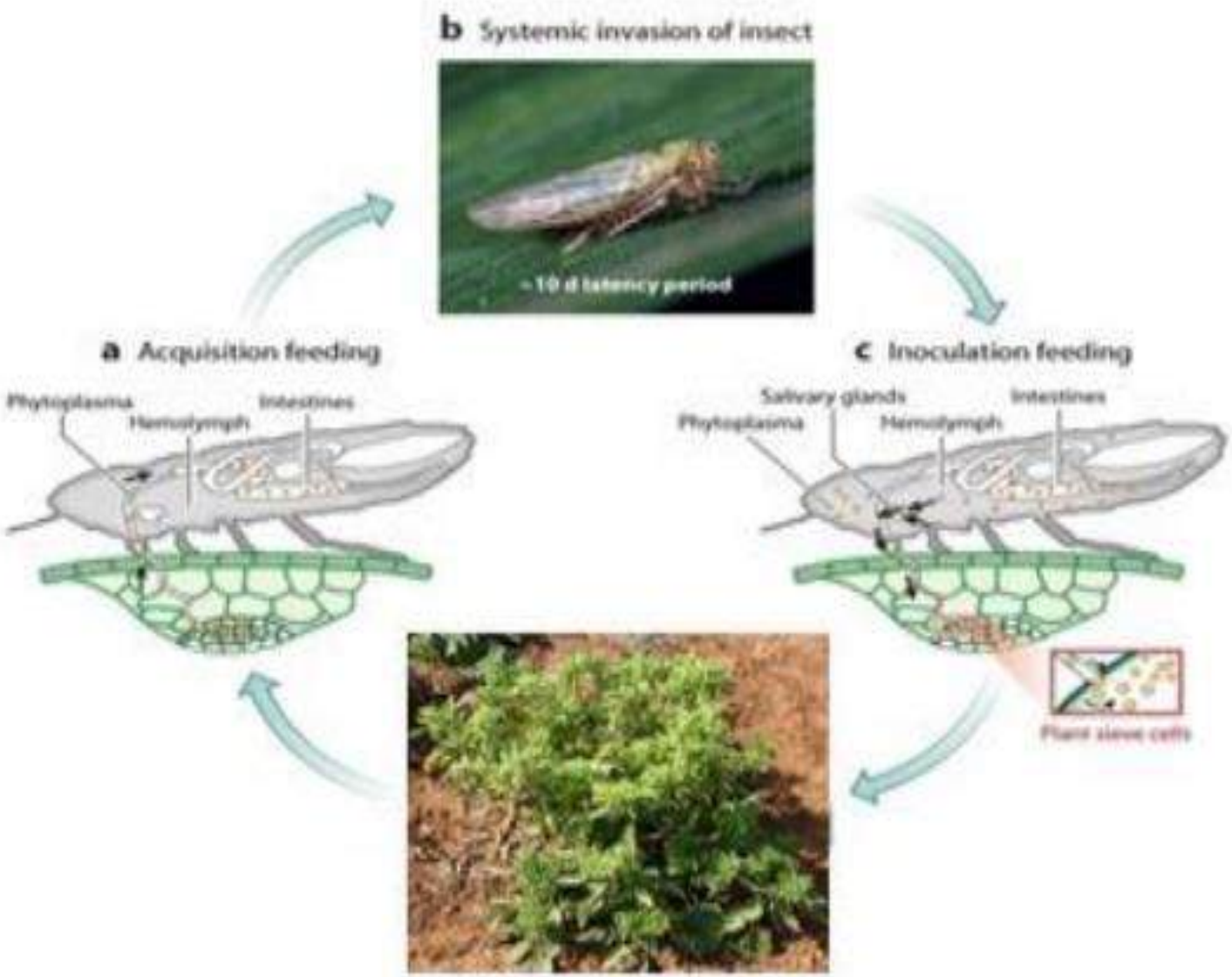
## Symptoms

- Infection is initially on one branch and later the entire plant shows symptoms
- **Reduction in size of newly formed leaves**
- Reduction in **petiole length** making the leaves appear to be sticking to the stem
- Affected plants have **narrow, soft, smooth and yellow** shorter leaves
- **Internodes of the stem are shortened**
- **Auxillary buds get** enlarged but their petioles and leaves also remain shortened giving the plant a bushy appearance

# Little leaf of brinjal – Phytoplasma

- When young plants are attacked no flowering and fruits
- No flowering but if flowers are formed they remain green
- Fruiting is rare, if any fruit is formed, it becomes **hard, tough and fails to mature**
- Young fruit **turns necrotic, get mummified**
- **Purple coloured fruits turn white after infection.**

# Little leaf of brinjal – Phytoplasma



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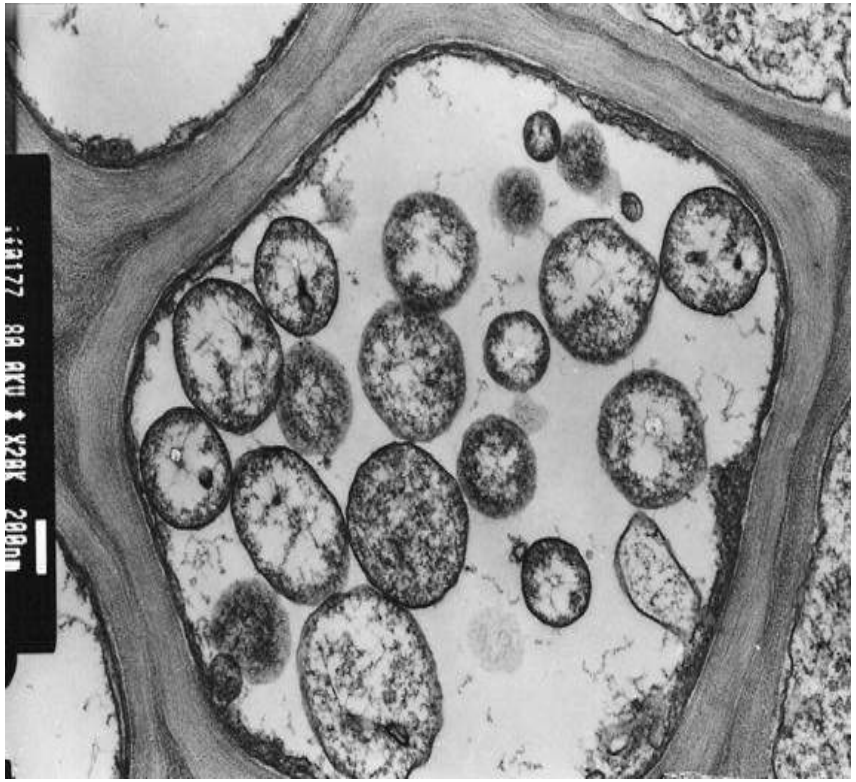


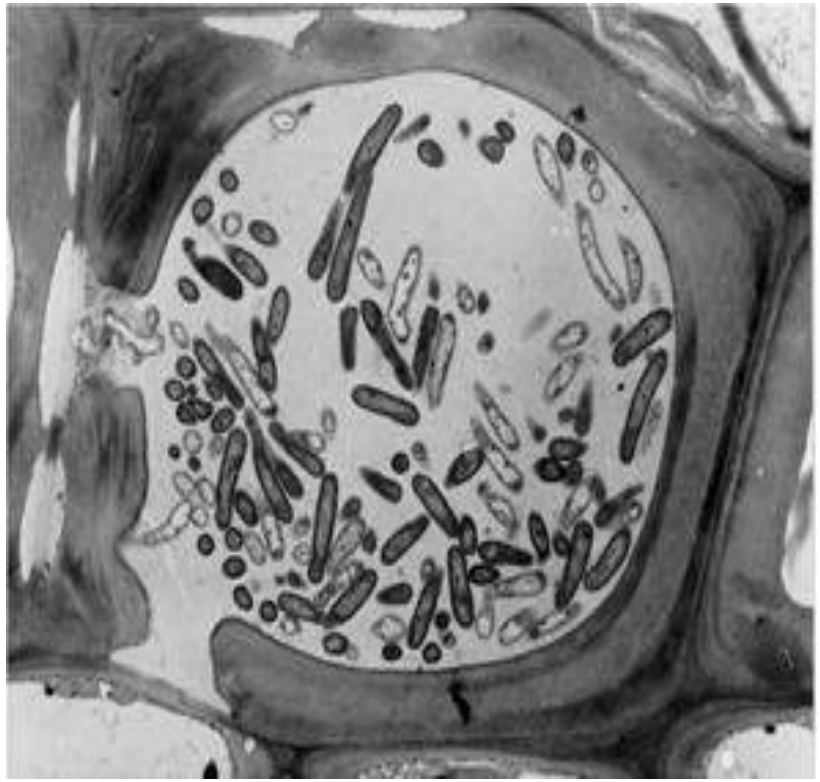

# Little leaf of brinjal – Phytoplasma

## Management

Moderately resist: Pusa Purple Round, Pusa purple cluster, Arka sheel, Manjari Gota, Banaras Gaint, Black beauty and Brinjal round

- Pant Rituraj, H-8,BB-7, BWR-12 are resistant to **little leaf, Phompsis blight and bacterial wilt**
- Destruction of affected plants
- Eradication of solanaceous weed hosts
- **Avoid planting of tomato and sweet potato** in neighbouring field as they increase the vector population
- **Early sowing in III week of June** helps in reducing the disease incidence

# Little leaf of brinjal – Phytoplasma

## Management contd..

- **Seed dip in tetracycline (10-50 ppm)**
- Dipping the seedlings in aqueous solution of tetracycline + Monocrotophos followed by soil application of Phorate 1 kg/ha
- Spray Methyl demeton or Dimethoate @ 2 ml/l or Imidacloprid @ 1.2 ml/l or soil application of Carbofuran granules
- By tying the cotton strip dipped in tetracycline to seedlings around the stem
- Spray Methyl demeton or Dimethoate @ 2 ml/l or Imidacloprid @ 1.2 ml/l or soil application of Carbofuran granules
- Spraying of Monocrotophos in nursery and main field at 10 days interval from the appearance of vectors atleast 10 times reduces the disease incidence